

第一周实验报告

一 课堂练习

1 问题一 “含空格文件名的批量整理” 实验过程：

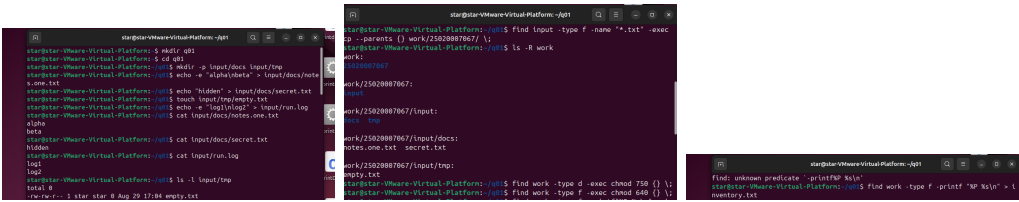


图 1: 实验过程截图

实验结果：

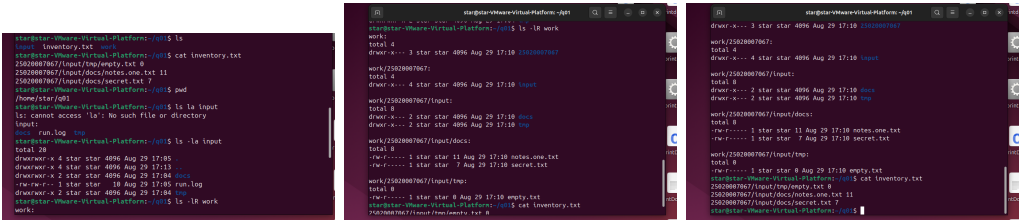
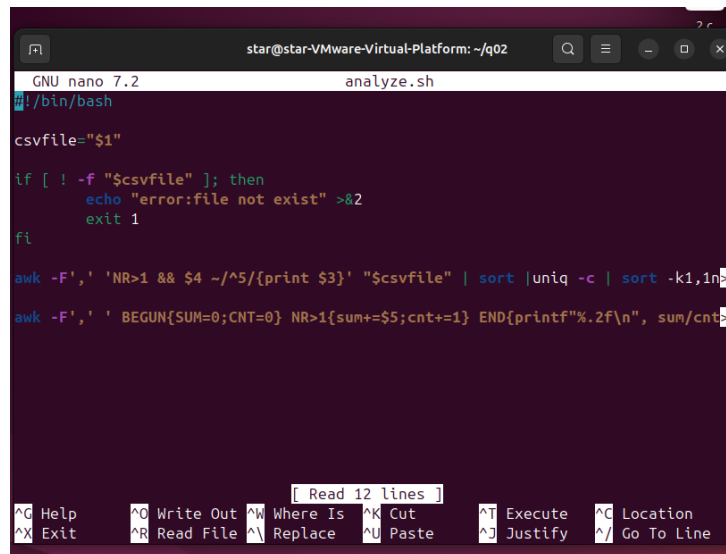


图 2: 实验结果截图

2 问题二 “随机访问日志统计” 实验过程：



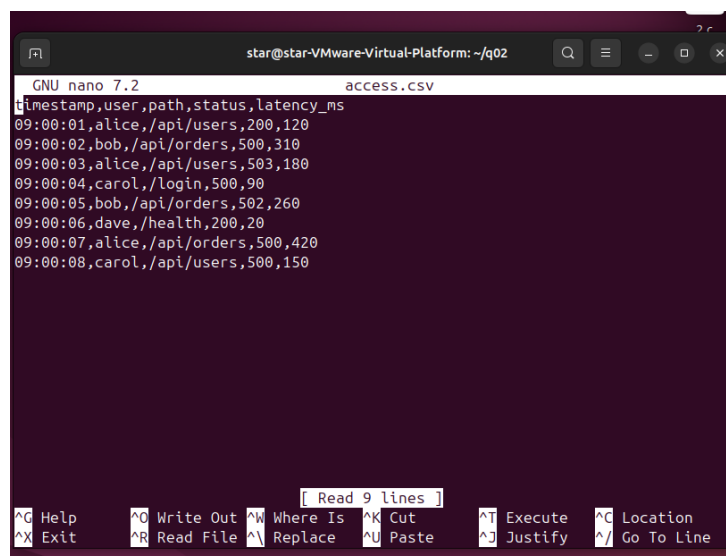
```
star@star-VMware-Virtual-Platform: ~/q02
GNU nano 7.2 analyze.sh
#!/bin/bash

csvfile="$1"

if [ ! -f "$csvfile" ]; then
    echo "error:file not exist" >&2
    exit 1
fi

awk -F',' 'NR>1 && $4 ~/^5/{print $3}' "$csvfile" | sort | uniq -c | sort -k1,1n
awk -F',' 'BEGIN{SUM=0;CNT=0} NR>1{sum+=5$5;cnt+=1} END{printf "%.2f\n", sum/cnt}'
```

图 3: 脚本代码



```
star@star-VMware-Virtual-Platform: ~/q02
GNU nano 7.2 access.csv
timestamp,user,path,status,latency_ms
09:00:01,alice,/api/users,200,120
09:00:02,bob,/api/orders,500,310
09:00:03,alice,/api/users,503,180
09:00:04,carol,/login,500,90
09:00:05,bob,/api/orders,502,260
09:00:06,dave,/health,200,20
09:00:07,alice,/api/orders,500,420
09:00:08,carol,/api/users,500,150
```

图 4: CSV文件内容

实验结果:

```
star@star-VMware-Virtual-Platform:~/q02$ ./analyze.sh access.csv
3 /api/orders
2 /api/users
193.75
star@star-VMware-Virtual-Platform:~/q02$ ./analyze.sh nofile.csv
error:file not exist
star@star-VMware-Virtual-Platform:~/q02$ echo $?
1
star@star-VMware-Virtual-Platform:~/q02$
```

图 5: 实验结果截图

3 问题三 “制造、解决并解释一次合并冲突” 实验过程:

```
star@star-VMware-Virtual-Platform:~/q03$ cd q03
star@star-VMware-Virtual-Platform:~/q03$ git log --oneline
96f50a2 (HEAD -> main) resolve merge conflict,keep safe and add note
ef2c433 (feature-b) set mode fast
49c540c (feature-a) set mode safe
6bd808b init config mode=normal
star@star-VMware-Virtual-Platform:~/q03$
```

图 6: 初始化仓库+main初始提交

```
star@star-VMware-Virtual-Platform: ~/q03
star@star-VMware-Virtual-Platform:~$ cd q03
star@star-VMware-Virtual-Platform:~/q03$ git log --oneline
96f50a2 (HEAD -> main) resolve merge conflict,keep safe and add note
ef2c433 (feature-b) set mode fast
49c540c (feature-a) set mode safe
6bd808b init config mode=normal
star@star-VMware-Virtual-Platform:~/q03$ git log --oneline --graph
* 96f50a2 (HEAD -> main) resolve merge conflict,keep safe and add note
| \
| * ef2c433 (feature-b) set mode fast
| * | 49c540c (feature-a) set mode safe
|/
* 6bd808b init config mode=normal
star@star-VMware-Virtual-Platform:~/q03$
```

图 7: feature-a分支提交

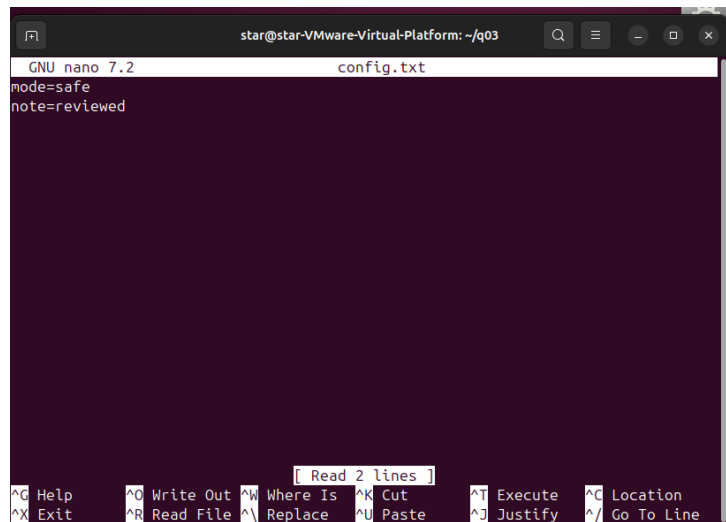
```
star@star-VMware-Virtual-Platform:~/q03$ git checkout feature-b
Switched to branch 'feature-b'
star@star-VMware-Virtual-Platform:~/q03$ git log --oneline
ef2c433 (HEAD -> feature-b) set mode fast
6bd808b init config mode=normal
star@star-VMware-Virtual-Platform:~/q03$
```

图 8: feature-b分支提交

合并冲突报错,执行git merge feature-b,终端输出Automatic merge failed;fix conflicts and then commit the result

原因:feature-a,feature-b在相同文件的同一行有不同修改;解决:保留mode=safe,新增note=reviewed,删除git冲突标记,add后完成合并提交

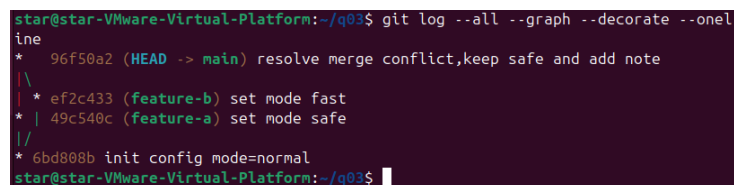
实验结果:



```
star@star-VMware-Virtual-Platform: ~/q03
GNU nano 7.2 config.txt
mode=safe
note=reviewed

[ Read 2 lines ]
^C Help  ^O Write Out  ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit  ^R Read File  ^L Replace  ^U Paste     ^J Justify   ^_ Go To Line
```

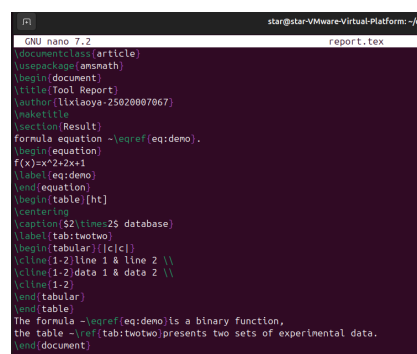
图 9: config.txt最终内容



```
star@star-VMware-Virtual-Platform: ~/q03$ git log --all --graph --decorate --oneline
* 96f50a2 (HEAD -> main) resolve merge conflict,keep safe and add note
|\
| * ef2c433 (feature-b) set mode fast
| * | 49c540c (feature-a) set mode safe
|/
* 6bd808b init config mode=normal
star@star-VMware-Virtual-Platform: ~/q03$
```

图 10: 图形日志

4 问题四 “修复并构建一页技术说明” 实验过程:



```
star@star-VMware-Virtual-Platform: ~/q03
GNU nano 7.2 report.tex
\documentclass{article}
\usepackage{amsmath}
\begin{document}
\title{Tool Report}
\author{lixiaoya-25820007867}
\maketitle
\section{Result}
formula equation -\eqref{eq:demo}.
\begin{equation}
f(x)=x^2+2x+1
\end{equation}
\label{eq:demo}
\end{equation}
\begin{table}[ht]
\caption{S2\times2S database}
\label{tab:twotwo}
\begin{tabular}{|c|c|}
\cline{1-2}\line 1 & line 2 \\\
\cline{1-2}data 1 & data 2 \\\
\cline{1-2}
\end{tabular}
\end{table}
The formula -\eqref{eq:demo}is a binary function,
the table -\ref{tab:twotwo}presents two sets of experimental data.
\end{document}
```

图 11: report.tex文件内容

```

star@star-VMware-Virtual-Platform: ~/q04$ latexmk -C report.tex
Rc files read:
/etc/LaTeXMk
Latexmk: This is Latexmk, John Collins, 31 Jan. 2024. Version 4.83.
Latexmk: Doing main (small) clean up for 'report.tex'

star@star-VMware-Virtual-Platform: ~/q04$ latexmk -pdf -halt-on-error report.tex
Rc files read:
/etc/LaTeXMk
Latexmk: This is Latexmk, John Collins, 31 Jan. 2024. Version 4.83.
No existing .aux file, so I'll make a simple one, and require run of *latex.
Latexmk: applying rule 'pdflatex'...
Rule 'pdflatex': Reasons for rerun
Category 'other':
  Rerun of 'pdflatex' forced or previously required:
    Reason or flag: 'Initial setup'

-----
Run number 1 of rule 'pdflatex'
-----
Running 'pdflatex -halt-on-error -recorder "report.tex"'
-----
This is pdfTeX, Version 3.141592653-2.6-1.40.25 (TeX Live 2023/Debian) (preloaded format=pdflatex)
restricted \write18 enabled.
entering extended mode

```

图 12: 指令运行

实验结果:

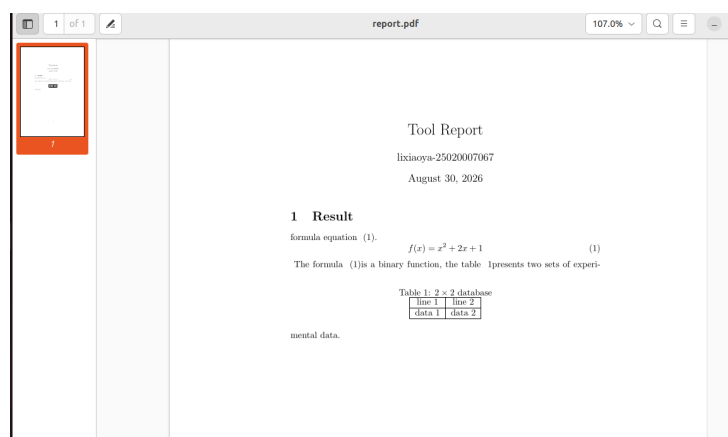


图 13: PDF文件内容

二 课后练习

- 1 问题一 ls 的 -l 选项(flag)作用是什么?运行 ls -l / 并观察输出.每一行最前面的 10 个字符分别代表什么?

作用: 显示文件的详细信息, 包括权限、所有者、文件大小和修改时间等.

```

star@star-VMware-Virtual-Platform:~$ ls -l
total 76
drwxr-xr-x 3 star star 4096 Aug 30 22:00 Desktop
drwxr-xr-x 2 star star 4096 Oct 21 2025 Documents
drwxr-xr-x 2 star star 4096 Oct 21 2025 Downloads
-rw-rw-r-- 1 star star 306 Oct 24 2025 fibonacci.go
-rw-rw-r-- 1 star star 404 Oct 26 2025 fibonacci.py
-rw-rw-r-- 1 star star 402 Oct 26 2025 fibonacci_recursive2.py
-rw-rw-r-- 1 star star 401 Oct 24 2025 fibonacci_recursive.go
-rw-rw-r-- 1 star star 477 Oct 26 2025 fibonacci_recursive.py
-rw-rw-r-- 1 star star 63 Oct 22 2025 hello.go
drwxrwxr-x 7 star star 4096 Aug 30 21:28 lab1
drwxr-xr-x 2 star star 4096 Oct 21 2025 Music
drwxr-xr-x 3 star star 4096 Nov 5 2025 Pictures
drwxr-xr-x 2 star star 4096 Oct 21 2025 Public
drwxrwxr-x 3 star star 4096 Aug 30 18:40 q03_backup
-rw-rw-r-- 1 star star 0 Aug 30 21:00 report.tex
drwx----- 6 star star 4096 Aug 24 14:58 snap
drwxr-xr-x 2 star star 4096 Oct 21 2025 Templates
-rw----- 1 star star 1 Oct 24 2025 test.go.save
-rw-rw-r-- 1 star star 173 Oct 29 2025 text.go
drwxr-xr-x 2 star star 4096 Oct 21 2025 Videos

```

图 14: 运行 `ls -l` / 输出

2 问题二三种引号的区别？写一条命令，输出一个同时包含三种特殊字符的字符串

区别:单引号保持原样输出,双引号和ANSI引号会进行变量替换和转义

```

star@star-VMware-Virtual-Platform:~$ echo '$\literal dollar: $ \nliteral exclamation:!'
literal dollar: $
literal exclamation:!!

```

图 15: 输出展示

3 问题三：写一个脚本，接收文件名参数（1），用 `test -f` 或 `[ -f ... ]` 检查该文件是否存在，并根据结果输出不同提示。

```

GNU nano 7.2                                check.sh
#!/bin/bash
if [ -f "$1" ]; then
    echo"文件$1存在"
else
    echo"文件不存在"
fi

```

图 16: 脚本代码

4 问题四把上一题完成的脚本保存为文件(如 check.sh).先运行 ./check.sh somefile,会发生什么?然后执行 chmod +x check.sh 再试一次.为什么这一步是必须的?

```
star@star-VMware-Virtual-Platform: $ nano check.sh
star@star-VMware-Virtual-Platform: $ ./check.sh somefile
bash: ./check.sh: Permission denied

star@star-VMware-Virtual-Platform: $ chmod +x check.sh
star@star-VMware-Virtual-Platform: $ ./check.sh somefile
./check.sh: line 5: echo文件不存在: command not found
star@star-VMware-Virtual-Platform: $
```

图 17: 指令运行

5 问题五在脚本的 set 选项(flag)里加入 -x 会发生什么?写个简单脚本试试并观察输出.

```
GNU nano 7.2 test.sh
#!/bin/bash
set -x

filename="note.txt"
echo "file name is $filename"
ls -l $filename

star@star-VMware-Virtual-Platform: $ chmod +x test.sh
+ chmod +x test.sh
star@star-VMware-Virtual-Platform: $ ./test.sh
+ ./test.sh
+ filename=note.txt
+ echo 'file name is note.txt'
file name is note.txt
+ ls -l note.txt
ls: cannot access 'note.txt': No such file or directory
```

图 18: 脚本及输出

6 问题六: 创建一个新的仓库,创建一个带有几行代码的文件

```
star@star-VMware-Virtual-Platform: $ cd recipe
star@star-VMware-Virtual-Platform:~/recipe$ git init
Initialized empty Git repository in /home/star/recipe/.git/
star@star-VMware-Virtual-Platform:~/recipe$ nano recipe.txt
star@star-VMware-Virtual-Platform:~/recipe$ git add recipe.txt
star@star-VMware-Virtual-Platform:~/recipe$ git commit -m "original recipe"
[main (root-commit) 8d811a3] original recipe
1 file changed, 3 insertions(+)
create mode 100644 recipe.txt
```

图 19: 创建仓库及文件

7 问题七: 提交后, 创建两个分支

```
star@star-VMware-Virtual-Platform:~/recipe$ git branch salty
star@star-VMware-Virtual-Platform:~/recipe$ git branch sweet
star@star-VMware-Virtual-Platform:~/recipe$ git branch
* main
  salty
  sweet
```

图 20: 创建分支

8 问题八：在分支中，修改一行

```
star@star-VMware-Virtual-Platform:~/recipe$ git checkout salty
Switched to branch 'salty'
star@star-VMware-Virtual-Platform:~/recipe$ nano recipe.txt
star@star-VMware-Virtual-Platform:~/recipe$ git add recipe.txt
star@star-VMware-Virtual-Platform:~/recipe$ git commit -m "changed to salt*2"
[salty f2f0072] changed to salt*2
1 file changed, 1 insertion(+), 1 deletion(-)
```

图 21: 修改分支内容

8 问题九：在分支中，对同一行进行不同修改

```
star@star-VMware-Virtual-Platform:~/recipe$ git checkout sweet
Switched to branch 'sweet'
star@star-VMware-Virtual-Platform:~/recipe$ nano recipe.txt
star@star-VMware-Virtual-Platform:~/recipe$ git add recipe.txt
star@star-VMware-Virtual-Platform:~/recipe$ git commit -m "changed to sugar*1"
[sweet d4d0f76] changed to sugar*1
1 file changed, 1 insertion(+), 1 deletion(-)
```

图 22: 修改分支内容

9 问题十：Now switch to and try , then . What happens? Look at the contents of - what do the , , and markers mean?

```
star@star-VMware-Virtual-Platform:~/recipe$ git checkout main
Switched to branch 'main'
star@star-VMware-Virtual-Platform:~/recipe$ git merge salty
Updating 8d811a3..f2f0072
Fast-forward
 recipe.txt | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
star@star-VMware-Virtual-Platform:~/recipe$ git merge sweet
Auto-merging recipe.txt
CONFLICT (content): Merge conflict in recipe.txt
Automatic merge failed; fix conflicts and then commit the result.
```

图 23: 提交结果

```
GNU nano 7.2          recipe.txt
<<<<<<< HEAD
salt*2
=====
sugar*1
>>>>>>> sweet
water*1
salt*1
```

图 24: 产生冲突后文件内容

三 GitHub提交

```
star@star-VMware-Virtual-Platform: ~/lab1
star@star-VMware-Virtual-Platform:~/lab1$ cd lab1
star@star-VMware-Virtual-Platform:~/lab1$
star@star-VMware-Virtual-Platform:~/lab1$ git log --all --graph --decorate --one
line
* 72c5f4a (HEAD -> main, origin/main) lab1:finish q04 latex report
* f557ebf lab1:finish q03 git merge conflict exercise
* 1f5e2f6 lab1:finish q02 git merge conflict exercise
* 14ef7bb lab1:finish q01 file batch arrange
```

图 25: commit记录截图

GitHub提交链接:<https://github.com/6-star-6/lab1>